	GLOBAL ECONOMICS **GRADE: 10TH** **TERM 2 – LESSON 3 PROJECT** **PROJECT PROBLEMS** **TEACHER'S NAME: Nicolás López Cuéllar**	**SECOND TERM 2025–2026**

Learning objective: Compare Bayes, Maximin, and Minimax strategies and establish the tree diagram for a decision-making problem.	**Assessment criteria:** C8: Compares Bayes, Maximin, and Minimax strategies. C9: Establishes the tree diagram for a decision-making problem.

Criteria assessment

Each assessment criterion is evaluated across the ten problems. A criterion is considered passed when it is correctly activated in at least 8 of the 10 problems.

1. Coffee-export logistics model under demand uncertainty

A coffee-export startup must choose one logistics model for the next season: A_1 (outsourced shipping), A_2 (mixed model), A_3 (own fleet), A_4 (regional consolidation hubs), or A_5 (strategic alliance network). Demand states and probabilities are $H : 0.45$, $M : 0.35$, and $L : 0.20$. Payoffs are in thousand USD. The payoff table is:

Payoff table

State	Prob.	A_1	A_2	A_3	A_4	A_5
H	0.45	150	142	165	155	148
M	0.35	98	112	90	118	104
L	0.20	52	78	35	70	88

Apply C8 by computing and ranking alternatives using Bayes (expected value), Maximin, and Minimax Regret, including any intermediate tables needed (EV summary and regret table). After obtaining the C10 summary and computing the C8 results, apply the following structured C9 decision procedure:

- Retain the top four alternatives according to EV (4 alternatives remain).
- From that subset, eliminate the alternative with the highest Max regret (3 alternatives remain).
- Select the alternative with the highest minimum payoff (Maximin) inside the reduced subset of three (final decision).

Using the C8 recommendation as the first-stage choice, apply C9 with second-stage uncertainty where the rule stays stable with probability 0.60 (payoff adjustment +12) or the rule tightens with probability 0.40 (payoff adjustment −18). Construct and evaluate the decision tree and state the final recommendation.

2. Textile production policy under market acceptance uncertainty

A textile manufacturer must choose one production policy: B_1 (premium fabrics), B_2 (balanced mix), B_3 (mass volume), B_4 (contract manufacturing), or B_5 (nearshoring hybrid). Market acceptance states and probabilities are $S_1 : 0.40$, $S_2 : 0.40$, and $S_3 : 0.20$. Payoffs are in thousand USD. The payoff table is:

Payoff table

State	Prob.	B_1	B_2	B_3	B_4	B_5
S_1	0.40	145	138	160	130	150
S_2	0.40	95	110	88	112	102
S_3	0.20	42	70	28	85	76

Apply C8 by computing and ranking alternatives using Bayes (expected value), Maximin, and Minimax Regret, including any intermediate tables needed (EV summary and regret table). After obtaining the C10 summary and computing the C8 results, apply the following structured C9 decision procedure:

- Keep the alternatives whose Max regret is at most 25 (2 alternatives remain).
- Between the remaining two alternatives, select the one with the higher EV (final decision).

Using the C8 recommendation as the first-stage choice, apply C9 with second-stage uncertainty where the audit passes with probability 0.70 (payoff adjustment +9) or the audit fails with probability 0.30 (payoff adjustment −21). Construct and evaluate the decision tree and state the final recommendation.

3. Platform launch strategy under enrollment uncertainty

A digital-learning company must choose one platform launch strategy: C_1 (fast launch), C_2 (guided rollout), C_3 (institutional partnership), C_4 (regional pilot clusters), or C_5 (franchise licensing). Enrollment states and probabilities are $E_1 : 0.30$, $E_2 : 0.45$, and $E_3 : 0.25$. Payoffs are in thousand USD. The payoff table is:

Payoff table

State	Prob.	C_1	C_2	C_3	C_4	C_5
E_1	0.30	168	150	158	162	148
E_2	0.45	92	118	106	114	120
E_3	0.25	30	84	74	80	96

Apply C8 by computing and ranking alternatives using Bayes (expected value), Maximin, and Minimax Regret, including any intermediate tables needed (EV summary and regret table). After obtaining the C10 summary and computing the C8 results, apply the following structured C9 decision procedure:

- Retain the top four alternatives according to EV (4 alternatives remain).
- Eliminate the alternative with the highest Max regret in that subset (3 alternatives remain).
- Retain the top two alternatives by Maximin from the reduced subset (2 alternatives remain).
- Use Minimax Regret as tie-break between the two finalists (final decision).

Using the C8 recommendation as the first-stage choice, apply C9 with second-stage uncertainty where there is a controlled incident with probability 0.75 (payoff adjustment +8) or a major outage with probability 0.25 (payoff adjustment −24). Construct and evaluate the decision tree and state the final recommendation.

4. Health-supply inventory strategy under demand-pressure uncertainty

A health-supply distributor must choose one inventory strategy: D_1 (aggressive stock), D_2 (balanced stock), D_3 (just-in-time), D_4 (regional micro-warehouses), or D_5 (vendor-managed inventory). Demand-pressure states and probabilities are $P_1 : 0.35$, $P_2 : 0.40$, and $P_3 : 0.25$. Payoffs are in thousand USD. The payoff table is:

Payoff table

State	Prob.	D_1	D_2	D_3	D_4	D_5
P_1	0.35	155	146	170	140	150
P_2	0.40	100	118	88	122	116
P_3	0.25	45	86	22	94	90

Apply C8 by computing and ranking alternatives using Bayes (expected value), Maximin, and Minimax Regret, including any intermediate tables needed (EV summary and regret table). After obtaining the C10 summary and computing the C8 results, apply the following structured C9 decision procedure:

- Retain the top three alternatives according to EV (3 alternatives remain).
- Keep only alternatives with Max regret at most 24 (2 alternatives remain).
- Choose the remaining alternative with the higher Maximin value among the two (final decision).

Using the C8 recommendation as the first-stage choice, apply C9 with second-stage uncertainty where fuel is stable with probability 0.55 (payoff adjustment +11) or there is a fuel spike with probability 0.45 (payoff adjustment −14). Construct and evaluate the decision tree and state the final recommendation.

5. Sustainable-packaging growth plan under demand uncertainty

A sustainable-packaging firm must choose one growth plan: E_1 (capacity expansion), E_2 (balanced portfolio), E_3 (high-volume contracts), E_4 (automation upgrade), or E_5 (strategic co-manufacturing). Demand scenarios and probabilities are $T_1 : 0.50$, $T_2 : 0.30$, and $T_3 : 0.20$. Payoffs are in thousand USD. The payoff table is:

Payoff table

State	Prob.	E_1	E_2	E_3	E_4	E_5
T_1	0.50	140	132	160	150	146
T_2	0.30	100	116	88	120	110
T_3	0.20	60	82	30	74	90

Apply C8 by computing and ranking alternatives using Bayes (expected value), Maximin, and Minimax Regret, including any intermediate tables needed (EV summary and regret table). After obtaining the C10 summary and computing the C8 results, apply the following structured C9 decision procedure:

- Retain the top three alternatives according to EV (3 alternatives remain).
- Remove the alternative with the lowest Maximax value inside that subset (2 alternatives remain).
- Select the alternative with the lower Max regret between the remaining two (final decision).

Using the C8 recommendation as the first-stage choice, apply C9 with second-stage uncertainty and construct/evaluate the decision tree to state the final recommendation.

6. Agri-processing commercialization plan under price-demand uncertainty

An agri-processing company must choose one commercialization plan: F_1 (export surge), F_2 (balanced contracts), F_3 (spot-market focus), F_4 (retail alliances), or F_5 (regional distributors). Price-demand states and probabilities are $R_1 : 0.25$, $R_2 : 0.50$, and $R_3 : 0.25$. Payoffs are in thousand USD. The payoff table is:

Payoff table

State	Prob.	F_1	F_2	F_3	F_4	F_5
R_1	0.25	170	150	162	148	156
R_2	0.50	96	118	90	124	112
R_3	0.25	40	86	28	92	80

Apply C8 by computing and ranking alternatives using Bayes (expected value), Maximin, and Minimax Regret, including any intermediate tables needed (EV summary and regret table). After obtaining the C10 summary and computing the C8 results, apply the following structured C9 decision procedure:

- Keep alternatives with minimum payoff at least 80 (3 alternatives remain).
- In that subset, eliminate the alternative with the highest Max regret (2 alternatives remain).
- Choose the remaining alternative with the higher EV between the two (final decision).

Using the C8 recommendation as the first-stage choice, apply C9 with second-stage uncertainty and construct/evaluate the decision tree to state the final recommendation.

7. Mobility-services expansion model under adoption uncertainty

A mobility-services startup must choose one expansion model: G_1 (rapid fleet growth), G_2 (balanced rollout), G_3 (premium-only focus), G_4 (urban hub model), or G_5 (partner network model). Adoption states and probabilities are $U_1 : 0.40$, $U_2 : 0.25$, and $U_3 : 0.35$. Payoffs are in thousand USD. The payoff table is:

Payoff table

State	Prob.	G_1	G_2	G_3	G_4	G_5
U_1	0.40	152	144	166	150	148
U_2	0.25	104	118	92	122	114
U_3	0.35	58	84	30	76	88

Apply C8 by computing and ranking alternatives using Bayes (expected value), Maximin, and Minimax Regret, including any intermediate tables needed (EV summary and regret table). After obtaining the C10 summary and computing the C8 results, apply the following structured C9 decision procedure:

- Retain the top three alternatives according to EV (3 alternatives remain).
- Retain the top two by Maximin from that subset (2 alternatives remain).
- Select the alternative with lower Max regret among the two finalists (final decision).

Using the C8 recommendation as the first-stage choice, apply C9 with second-stage uncertainty and construct/evaluate the decision tree to state the final recommendation.

8. Renewable-energy market-entry package under revenue uncertainty

A renewable-energy integrator must choose one market-entry package: H_1 (large utility bids), H_2 (balanced project mix), H_3 (merchant exposure), H_4 (storage-first strategy), or H_5 (distributed generation network). Revenue states and probabilities are $V_1 : 0.30$, $V_2 : 0.30$, and $V_3 : 0.40$. Payoffs are in thousand USD. The payoff table is:

Payoff table

State	Prob.	H_1	H_2	H_3	H_4	H_5
V_1	0.30	160	146	172	154	150
V_2	0.30	108	120	90	126	116
V_3	0.40	52	86	34	78	92

Apply C8 by computing and ranking alternatives using Bayes (expected value), Maximin, and Minimax Regret, including any intermediate tables needed (EV summary and regret table). After obtaining the C10 summary and computing the C8 results, apply the following structured C9 decision procedure:

- Retain the two alternatives with the lowest Max regret (2 alternatives remain).
- Between these two finalists, choose the one with the higher minimum payoff (Maximin) (final decision).

Using the C8 recommendation as the first-stage choice, apply C9 with second-stage uncertainty and construct/evaluate the decision tree to state the final recommendation.

9. Food-delivery operational strategy under demand uncertainty

A food-delivery platform must choose one operational strategy: I_1 (rapid-area expansion), I_2 (balanced city focus), I_3 (premium-only channels), I_4 (dark-kitchen scaling), or I_5 (partner franchise mesh). Demand states and probabilities are W_1 : 0.20, W_2 : 0.50, and W_3 : 0.30. Payoffs are in thousand USD. The payoff table is:

Payoff table

State	Prob.	I_1	I_2	I_3	I_4	I_5
W_1	0.20	148	142	164	156	150
W_2	0.50	110	124	94	118	112
W_3	0.30	60	88	26	82	96

Apply C8 by computing and ranking alternatives using Bayes (expected value), Maximin, and Minimax Regret, including any intermediate tables needed (EV summary and regret table). After obtaining the C10 summary and computing the C8 results, apply the following structured C9 decision procedure:

- Retain the three best alternatives by EV (including the tie at 114.80) (3 alternatives remain).
- Keep alternatives with Max regret not exceeding 20 (2 alternatives remain).
- Compare EV inside the reduced subset; since EV is tied, keep both alternatives (2 alternatives remain).
- Break the tie using Maximin between the two finalists (final decision).

Using the C8 recommendation as the first-stage choice, apply C9 with second-stage uncertainty and construct/evaluate the decision tree to state the final recommendation.

10. Cloud-services enterprise strategy under market uncertainty

A cloud-services provider must choose one enterprise strategy: J_1 (high-growth acquisition), J_2 (balanced subscriptions), J_3 (aggressive discounting), J_4 (sector-specialized bundles), or J_5 (channel-partner ecosystem). Market states and probabilities are Z_1 : 0.45, Z_2 : 0.25, and Z_3 : 0.30. Payoffs are in thousand USD. The payoff table is:

Payoff table

State	Prob.	J_1	J_2	J_3	J_4	J_5
Z_1	0.45	158	146	170	152	148
Z_2	0.25	102	118	88	124	112
Z_3	0.30	54	90	20	80	94

Apply C8 by computing and ranking alternatives using Bayes (expected value), Maximin, and Minimax Regret, including any intermediate tables needed (EV summary and regret table). After obtaining the C10 summary and computing the C8 results, apply the following structured C9 decision procedure:

- Retain the top three alternatives according to EV (3 alternatives remain).
- Eliminate the alternative with the highest Max regret within that subset (2 alternatives remain).
- Select the alternative with the highest minimum payoff (Maximin) from the two finalists (final decision).

Using the C8 recommendation as the first-stage choice, apply C9 with second-stage uncertainty and construct/evaluate the decision tree to state the final recommendation.